STRUCTURAL DRAWING
FOR
ONE STOREYED
RCC RESIDENTIAL BUILDING
(NAYPYITAW)

DRAWING LIST

SR. NO.	SHEET NO.	DRAWING TITLE	REV.
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9.	SD09	Column Layout Plan	
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12.	SD12	Roof Floor Beam Layout Plan	
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OWNER:

PROJECT:
PROPOSED ONE STOREYED
R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:

SCALE	AS SHOWN
DATE	

SHEET NO.

RSE (STRUCTURE):

RSE (CONSTRUCTION):

OWNER:

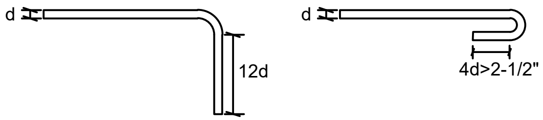
DESIGN SPECIFICATION

CONCRETE CYLINDER STRENGTH - 2500 PSI
REBAR YIELD STRENGTH - 50000 PSI
STRUCTURAL STEEL YIELD STRENGTH - 36000 PSI

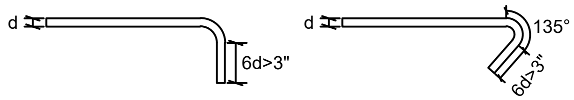
COVER

- FOOTING - 3"
- COLUMN - 1.5"
- BEAM - 1.5"
- SLAB - 0.75"

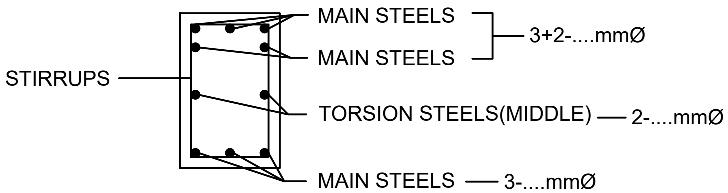
STANDARD HOOK (MAIN BARS)



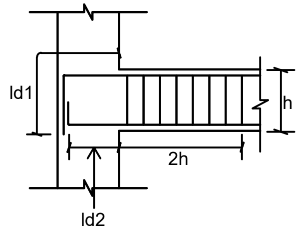
STANDARD HOOK (STIRRUPS & TIES)



REBAR POSITION DETAIL



TYPICAL BEAM ANCHORAGE



NOTE - Beam anchorage length (Ld1) should be used for both top and bottom rebars in beam-column joints.

OWNER:

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PROPOSED ONE STOREYED R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
STRUCTURAL DETAIL 1

SCALE	AS SHOWN
DATE	

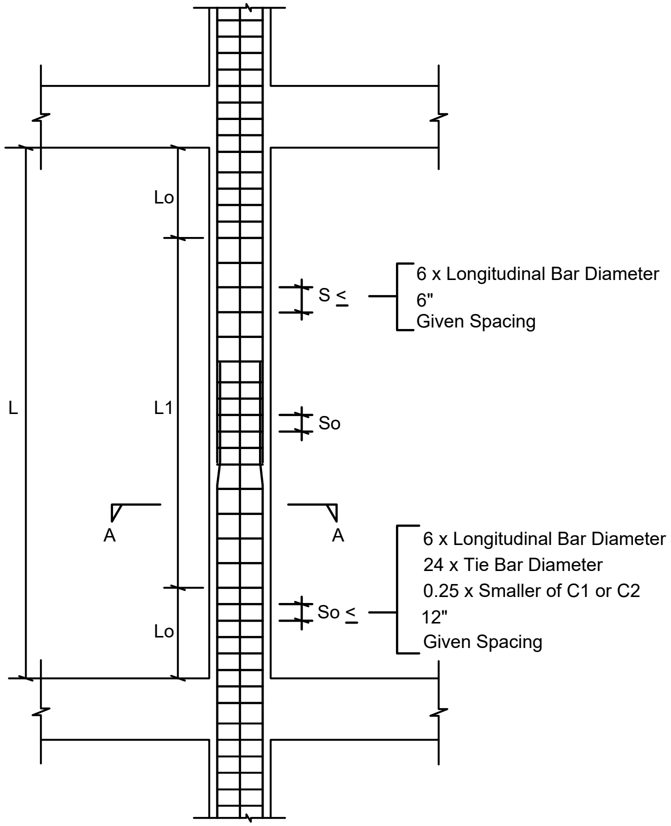
SHEET NO.
SD01

RSE (STRUCTURE):

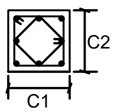
RSE (CONSTRUCTION):

OWNER:

TYPICAL COLUMN DETAILING

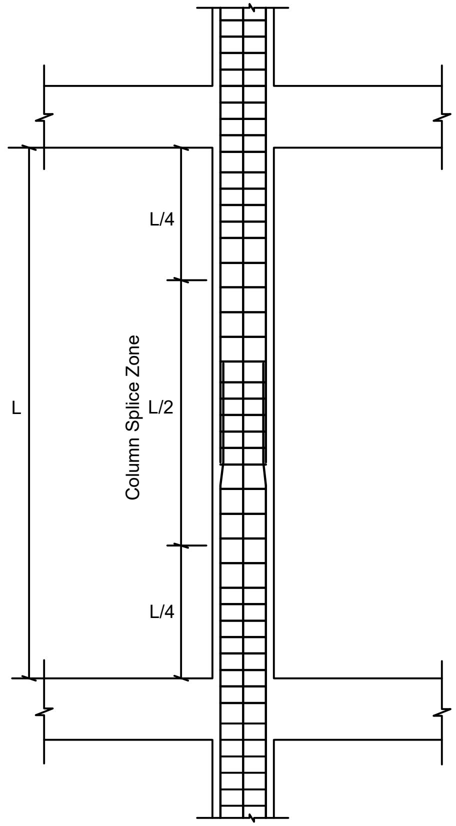


Lo ≥
- Larger of C1 or C2
- Clear Height / 6
- 18"



SECTION A-A

COLUMN SPLICE DETAILING



NOTE - Column clear cover is 1.5".
- Do not splice not more than 50% of total steel area at any section.
- Column may be spliced within center half of column clear height .
- Column rebar splice length should be tension splice length.

OWNER:

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PROPOSED ONE STOREYED R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
STRUCTURAL DETAIL 2

SCALE	AS SHOWN
DATE	

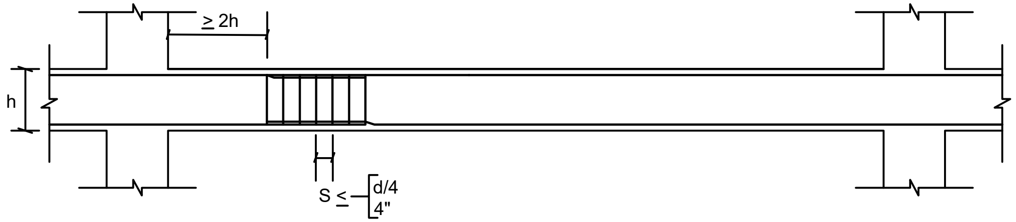
SHEET NO.
SD02

RSE (STRUCTURE):

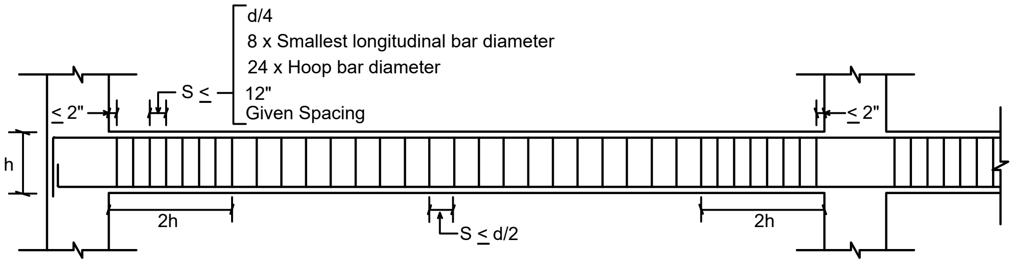
RSE (CONSTRUCTION):

OWNER:

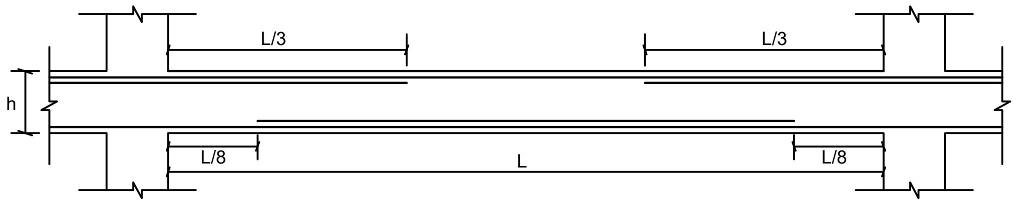
TYPICAL BEAM SPLICE DETAIL



TYPICAL BEAM STIRRUPS DETAIL



STEEL CUT-OFF POINT FOR BEAM



NOTE - Beam clear cover is 1.5".
- Do not splice not more than 50% of total steel area at any section.
- Beam rebar lap splice are away from potential hinge zone.

OWNER:

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BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
STRUCTURAL DETAIL 3

SCALE	AS SHOWN
DATE	

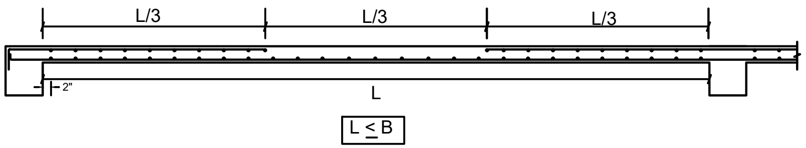
SHEET NO.
SD03

RSE (STRUCTURE):

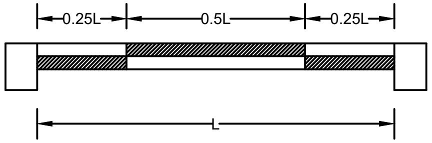
RSE (CONSTRUCTION):

OWNER:

TYPICAL SLAB DETAILING

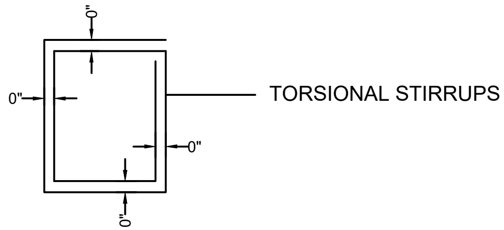


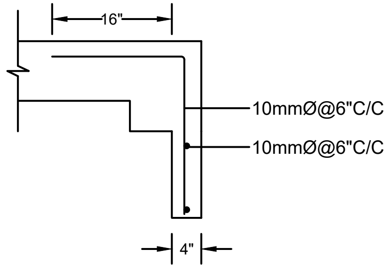
LAP LOCATION



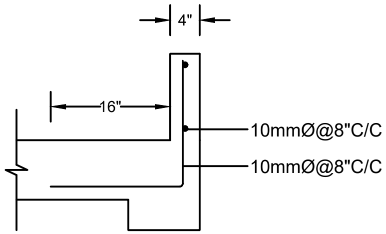
NOTE - Slab clear cover is 0.75".

- Slab rebar splice length should be beam splice length.
- Slab rebar splice location should be staggered .

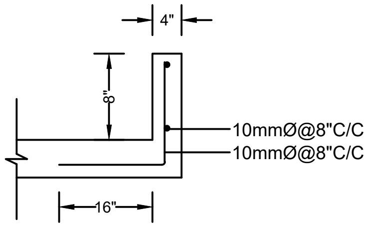




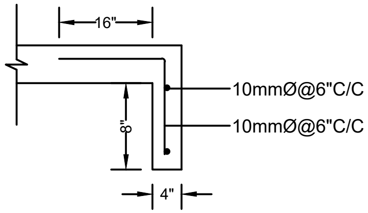
DOWNWALL FROM BEAM



UPSTAND FROM BEAM



UPSTAND FROM SLAB



DOWNWALL FROM SLAB

OWNER:

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PROPOSED ONE STOREYED
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BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
STRUCTURAL DETAIL 4

SCALE	AS SHOWN
DATE	

SHEET NO.
SD04

RSE (STRUCTURE):

RSE (CONSTRUCTION):

OWNER:

TENSION DEVELOPMENT LENGTH, Ld (in)

Bar Size (mm)	f'c=2.5ksi			f'c=3.0ksi			f'c=4.0ksi		
	fy (ksi)			fy (ksi)			fy (ksi)		
	40	50	60	40	50	60	40	50	60
10	13	16	19	12	15	18	12	13	15
12	16	19	23	14	18	21	12	15	18
16	21	26	31	19	23	28	16	20	24
18	23	29	34	21	26	31	18	23	27
20	26	32	38	23	29	35	20	25	30

COMPRESSION DEVELOPMENT LENGTH, Ld (in)

Bar Size (mm)	f'c=2.5ksi			f'c=3.0ksi			f'c=4.0ksi		
	fy (ksi)			fy (ksi)			fy (ksi)		
	40	50	60	40	50	60	40	50	60
10	8	8	10	8	8	9	8	8	8
12	8	10	12	8	9	11	8	8	9
16	10	13	16	10	12	14	8	10	12
18	12	15	17	11	13	16	9	12	14
20	13	16	19	12	15	18	10	13	15

TENSION SPLICE LENGTH (in) (1.0 Ld)

Bar Size (mm)	f'c=2.5ksi			f'c=3.0ksi			f'c=4.0ksi		
	fy (ksi)			fy (ksi)			fy (ksi)		
	40	50	60	40	50	60	40	50	60
10	13	16	19	12	15	18	12	13	15
12	16	19	23	14	18	21	12	15	18
16	21	26	31	19	23	28	16	20	24
18	23	29	34	21	26	31	18	23	27
20	26	32	38	23	29	35	20	25	30

TENSION SPLICE LENGTH (in) (1.3 Ld)

Bar Size (mm)	f'c=2.5ksi			f'c=3.0ksi			f'c=4.0ksi		
	fy (ksi)			fy (ksi)			fy (ksi)		
	40	50	60	40	50	60	40	50	60
10	17	21	25	16	20	24	16	17	20
12	21	25	30	19	24	28	16	20	24
16	28	34	41	25	30	37	21	26	32
18	30	38	45	28	34	41	24	30	36
20	34	42	50	30	38	46	26	33	39

COMPRESSION SPLICE LENGTH (in)

Bar Size (mm)	f'c=2.5ksi			f'c>=3.0ksi		
	fy (ksi)			fy (ksi)		
	40	50	60	40	50	60
10	12	13	16	12	12	12
12	13	16	19	12	12	15
16	17	21	26	13	16	19
18	19	24	29	15	18	22
20	21	27	32	16	20	24

ANCHORAGE LENGTH (in)

Bar Size (mm)	f'c=2.5ksi						f'c=3.0ksi					
	fy (ksi)						fy (ksi)					
	40		50		60		40		50		60	
	ld1	ld2	ld1	ld2	ld1	ld2	ld1	ld2	ld1	ld2	ld1	ld2
10	12	>=6	14	>=6	15	>=6	12	>=6	13	>=6	14	>=6
12	15	>=6	16	>=6	18	>=6	14	>=6	16	>=6	17	>=6
16	19	>=6	22	>=6	24	>=6	18	>=6	21	>=6	23	>=6
18	22	>=6	25	>=6	27	>=6	21	>=6	23	>=6	26	>=6
20	24	>=6	27	>=6	30	>=6	23	>=6	26	>=6	29	>=6

OWNER:

PROJECT:

PROPOSED ONE STOREYED
R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:

STRUCTURAL DETAIL 5

SCALE	AS SHOWN
DATE	

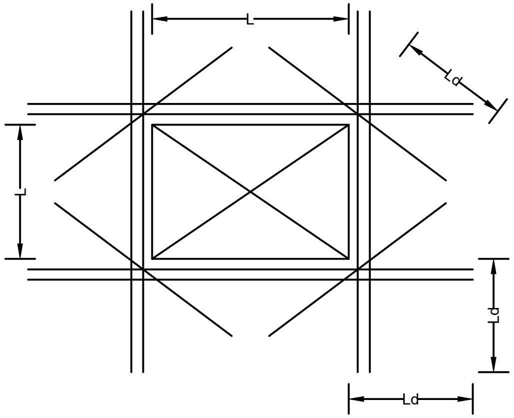
SHEET NO.

SD05

RSE (STRUCTURE):

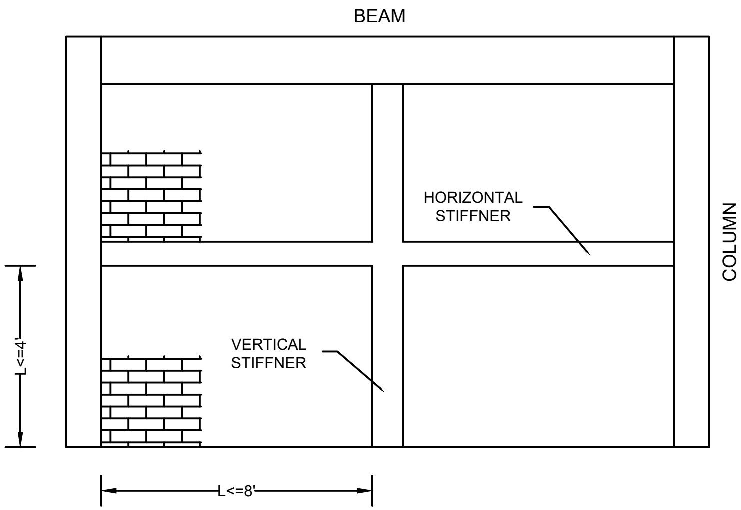
RSE (CONSTRUCTION):

OWNER:

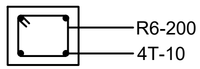


FLOOR OPENING (L)	ADD BAR
L<10"	2-10mmØ
10"<=L<20"	2-12mmØ
20"<=L<40"	2-16mmØ

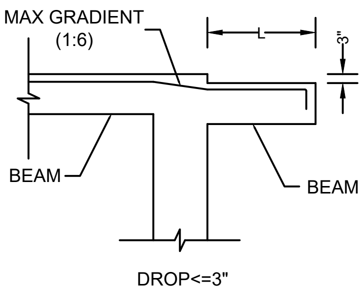
FLOOR OPENING DETAIL

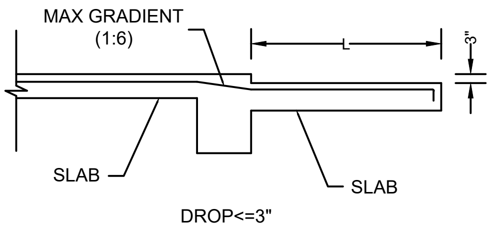


BRICKWALL STIFFNER DETAIL



STIFFNER SECTION





OWNER:

PROJECT:
PROPOSED ONE STOREYED
R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
STRUCTURAL DETAIL 6

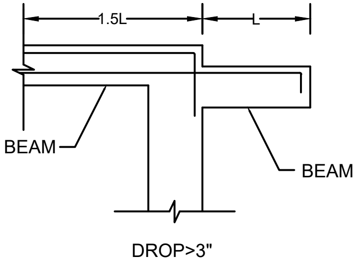
SCALE	AS SHOWN
DATE	

SHEET NO.
SD06

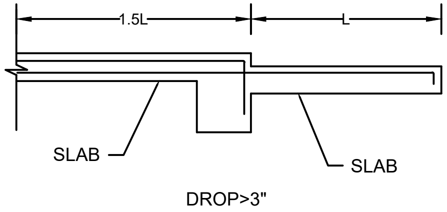
RSE (STRUCTURE):

RSE (CONSTRUCTION):

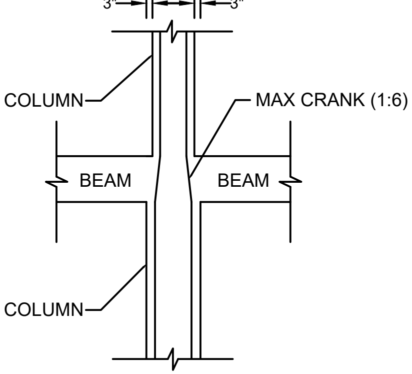
OWNER:



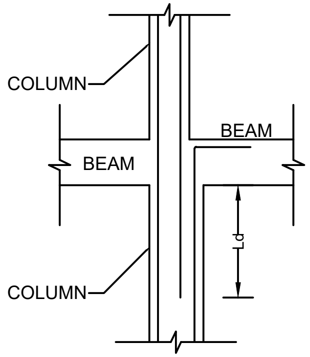
BEAM DROP DETAIL



SLAB DROP DETAIL



REDUCED SIZE<=3"



REDUCED SIZE>3"

COLUMN SIZE REDUCE DETAIL

OWNER:

PROJECT:

PROPOSED ONE STOREYED
R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:

STRUCTURAL DETAIL 7

SCALE	AS SHOWN
DATE	

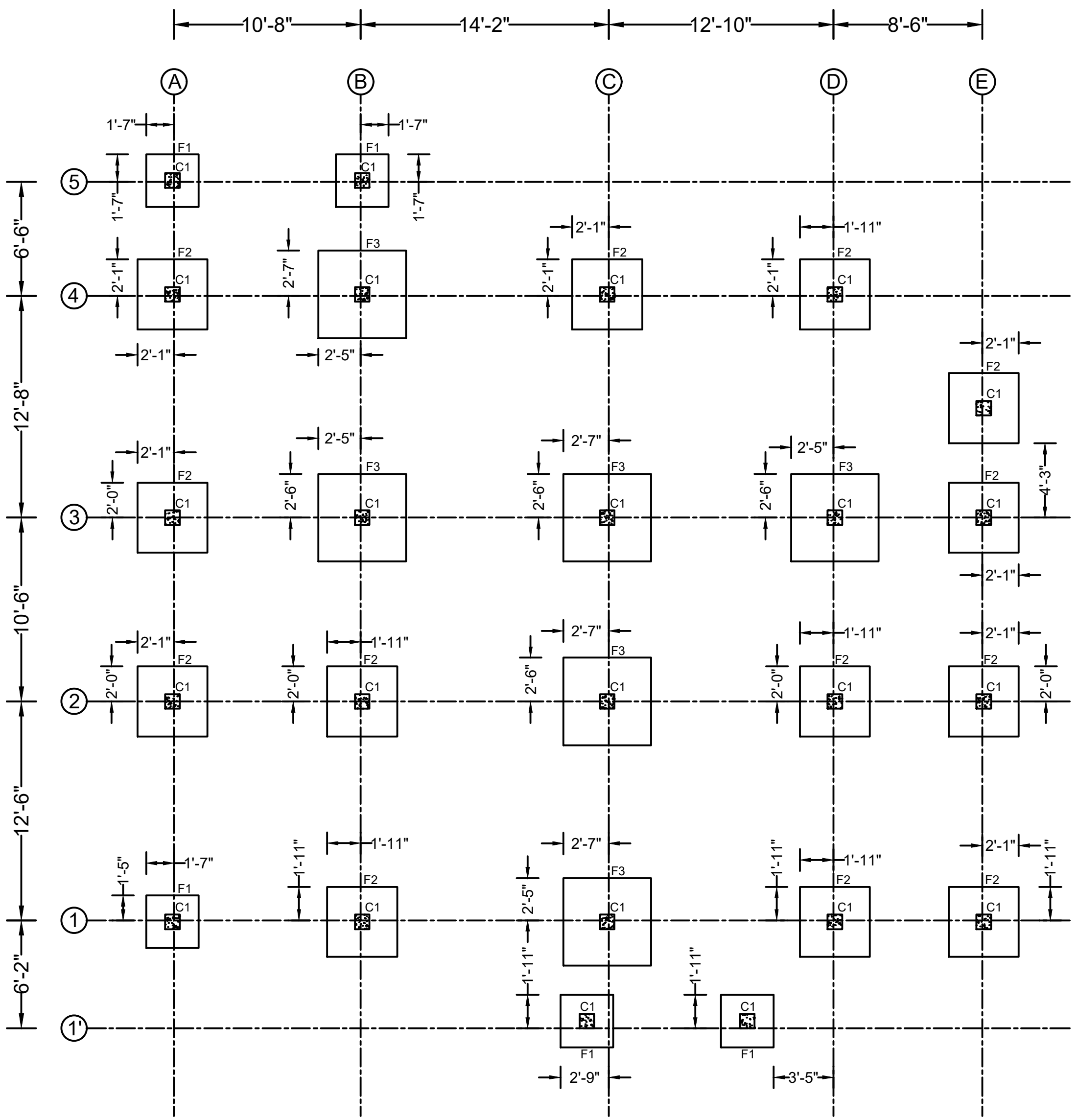
SHEET NO.

SD07

RSE (STRUCTURE):

RSE (CONSTRUCTION):

OWNER:



LEGEND

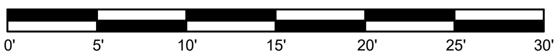
F1	3'-0"x3'-0"x0'-10"
F2	4'-0"x4'-0"x0'-10"
F3	5'-0"x5'-0"x0'-10"

FOOTING LAYOUT PLAN

Note

-Footing & column location should be reviewed & checked with architectural drawing.
-Footing clear cover is 3".
-Assumed soil allowable bearing capacity is 0.8 tons/sqft.

Scale :



OWNER:

PROJECT:
PROPOSED ONE STOREYED
R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
FOOTING LAYOUT PLAN

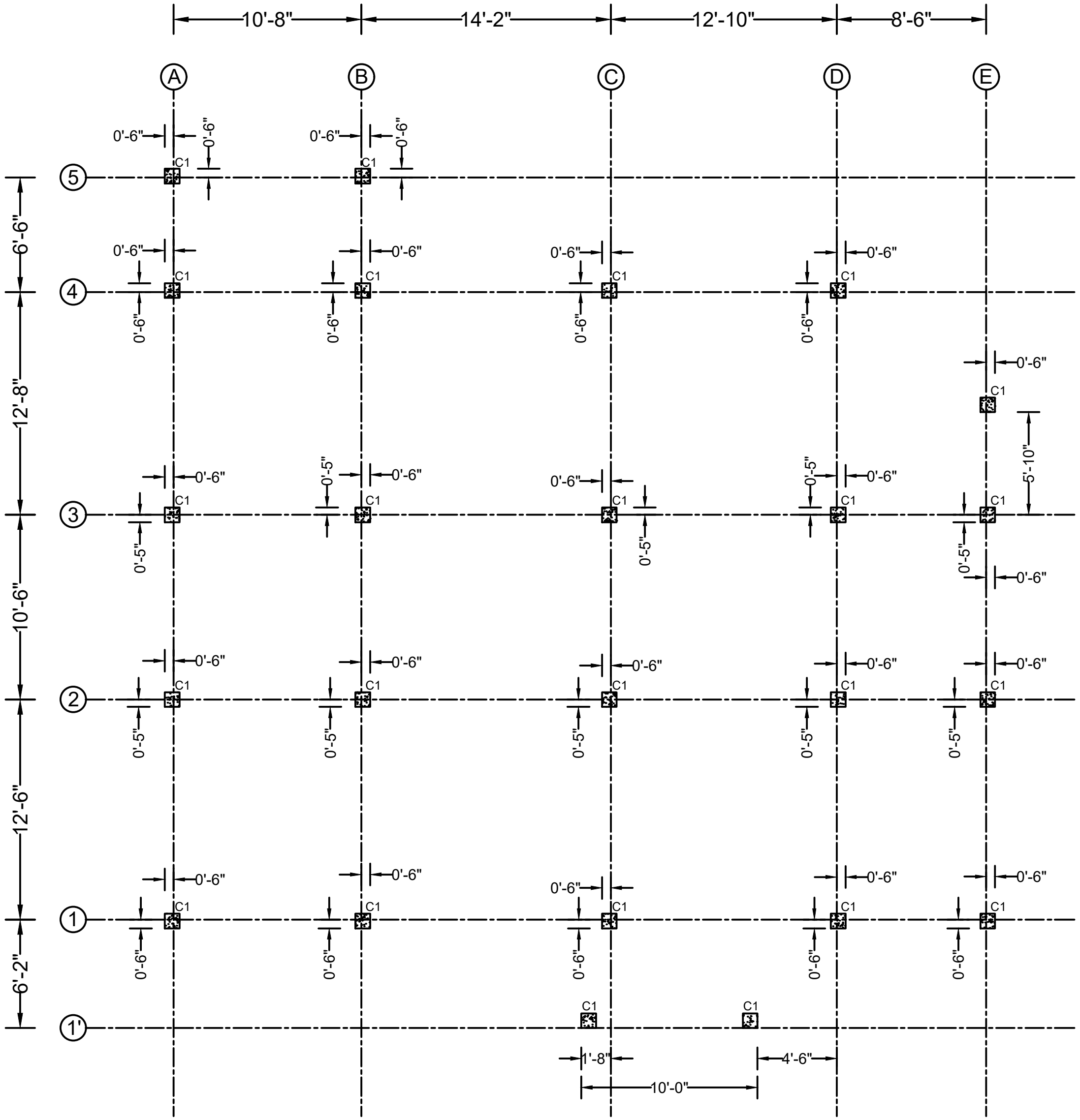
SCALE	AS SHOWN
DATE	

SHEET NO.
SD08

RSE (STRUCTURE):

RSE (CONSTRUCTION):

OWNER:



COLUMN LAYOUT PLAN

Scale :

Note

-Column clear cover is 1.5".
-Column rebars should not be spliced full bars in one section.
-Columns splice length should be tension splice length.
-Ties seismic (135°) hook must be alternative.

OWNER:

PROJECT:

PROPOSED ONE STOREYED R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:

COLUMN LAYOUT PLAN

SCALE	AS SHOWN
DATE	

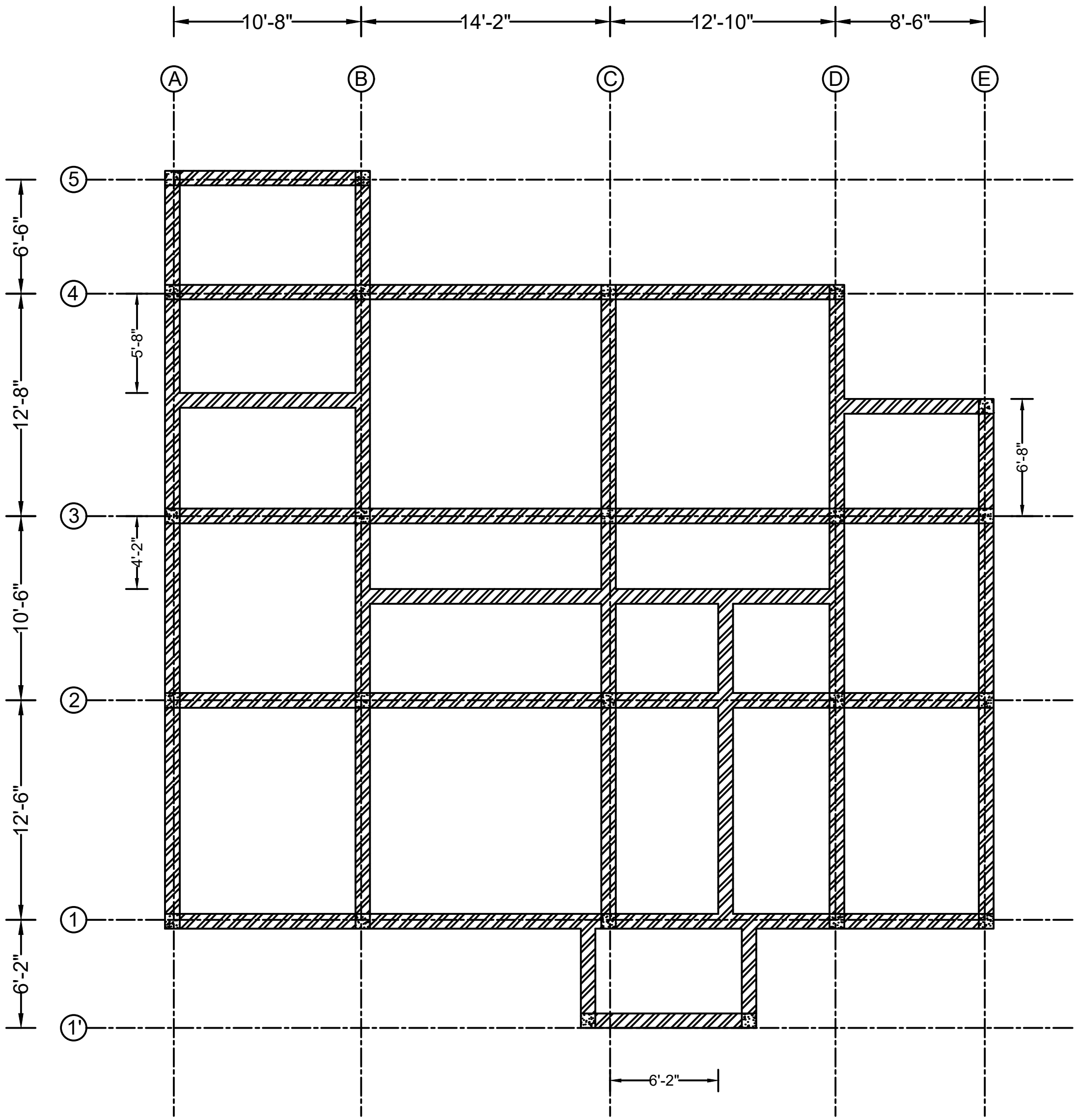
SHEET NO.

SD09

RSE (STRUCTURE):

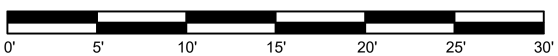
RSE (CONSTRUCTION):

OWNER:

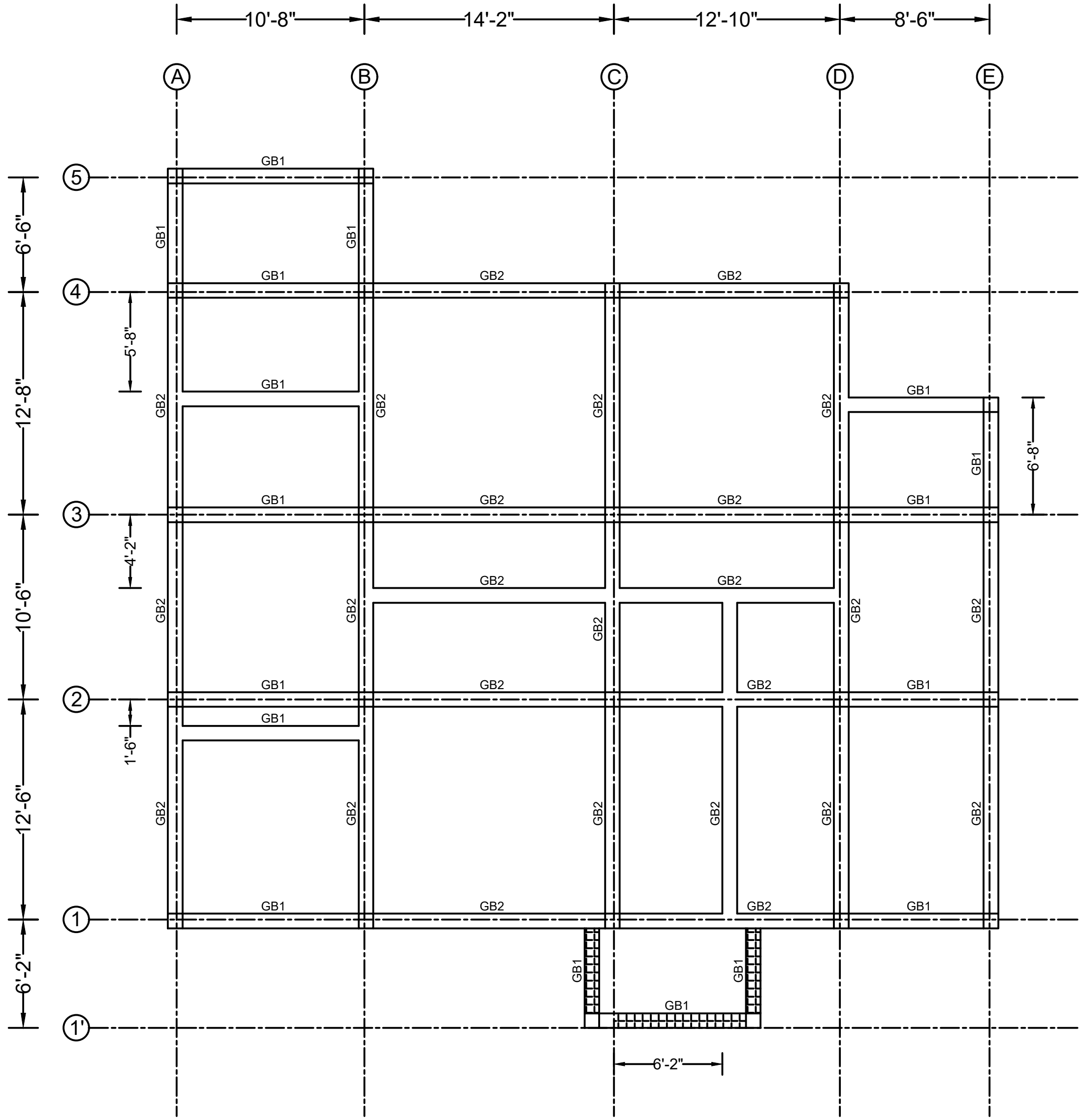


COLUMN LAYOUT PLAN

Scale :



OWNER:	
PROJECT:	
BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW
SUBJECT: RETAINING WALL PLAN	
SCALE	
DATE	
SHEET NO. SD10	
RSE (STRUCTURE):	
RSE (CONSTRUCTION):	
OWNER:	



GROUND FLOOR BEAM LAYOUT PLAN

LEGEND

GB1	10"x10"
GB2	10"x12"

6" Down

Column size reduce and end at this level.

Note
-Beam offset should be checked with architectural brick wall drawing.
-Beam clear cover is 1.5".
-Beams should not be spliced full bars in one section.
-Stirrups seismic (135°) hook must be alternative.
-Beam & slab level should be architectural & site condition.

Scale :

0' 5' 10' 15' 20' 25' 30'

OWNER:

PROJECT:
PROPOSED ONE STOREYED R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
GROUND FLOOR BEAM LAYOUT PLAN

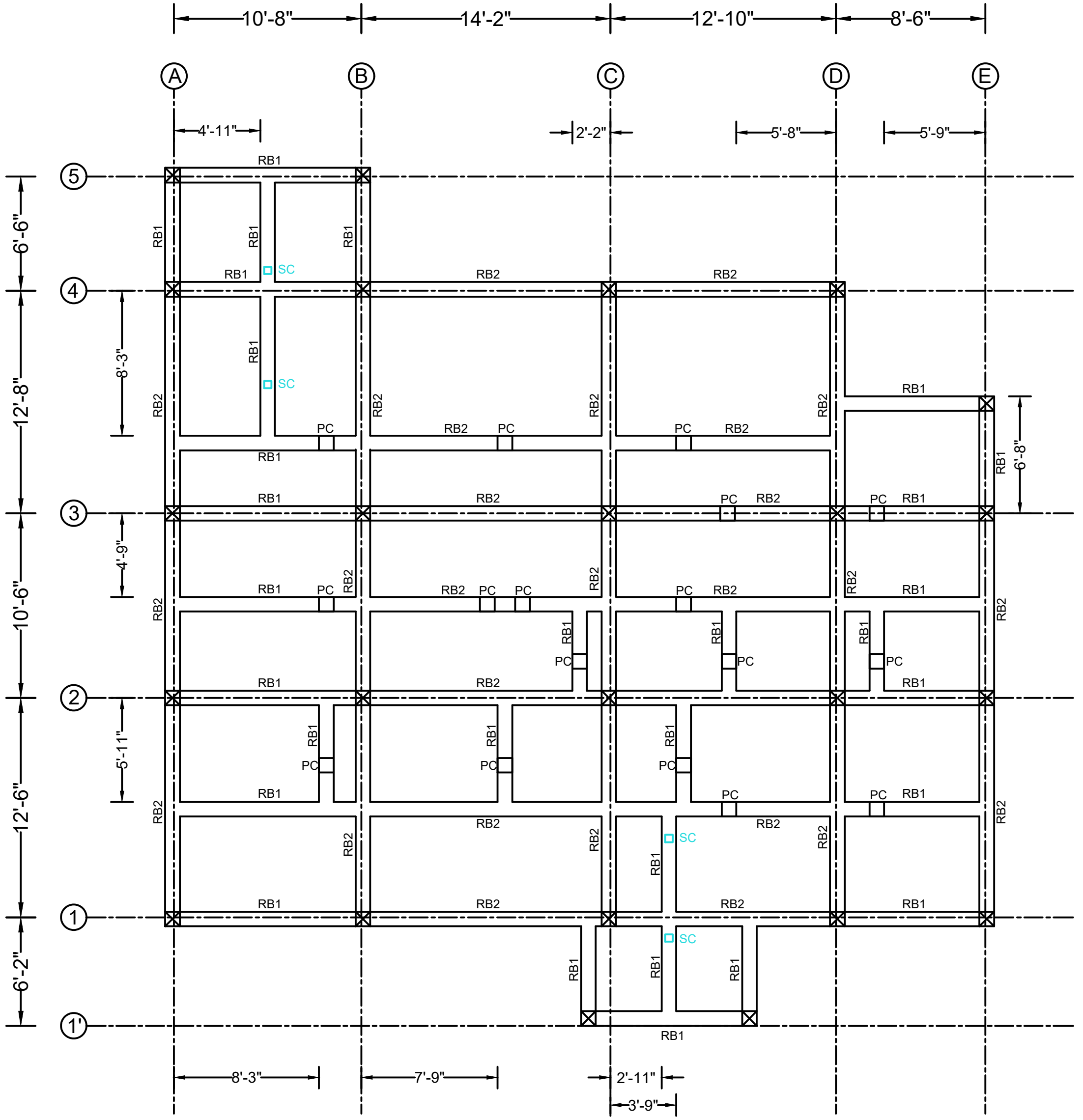
SCALE	AS SHOWN
DATE	

SHEET NO.
SD11

RSE (STRUCTURE):

RSE (CONSTRUCTION):

OWNER:



ROOF FLOOR BEAM LAYOUT PLAN

LEGEND

RB1	10"x10"
RB2	10"x12"

Column size reduce and end at this level.

Note
-Beam offset should be checked with architectural brick wall drawing.
-Beam clear cover is 1.5".
-Beams should not be spliced full bars in one section.
-Stirrups seismic (135°) hook must be alternative.
-Beam & slab level should be architectural & site condition.
-SC is the steel column member that is 5"x5"(1.4mm thk;).

Scale :

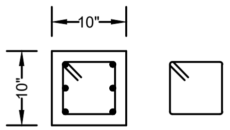
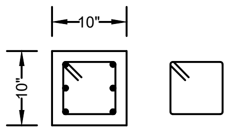
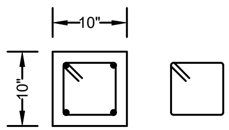
0' 5' 10' 15' 20' 25' 30'

OWNER:	
PROJECT: PROPOSED ONE STOREYED R.C.C BUILDING	
BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW
SUBJECT: ROOF FLOOR BEAM LAYOUT PLAN	
SCALE	AS SHOWN
DATE	
SHEET NO. SD12	
RSE (STRUCTURE):	
RSE (CONSTRUCTION):	
OWNER:	

FOOTING SCHEDULE

TYPE	SIZE			DEPTH	BOTTOM REINFORCEMENT		TOP REINFORCEMENT		REMARKS
	L	B	H		X-DIRECTION	Y-DIRECTION	X-DIRECTION	Y-DIRECTION	
F1	3'-0"	3'-0"	0'-10"	4ft below NGL	16mmØ@6"C/C	16mmØ@6"C/C	—	—	Isolated
F2	4'-0"	4'-0"	0'-10"	4ft below NGL	16mmØ@6"C/C	16mmØ@6"C/C	—	—	Isolated
F3	5'-0"	5'-0"	0'-10"	4ft below NGL	16mmØ@6"C/C	16mmØ@6"C/C	—	—	Isolated

COLUMN SCHEDULE

TYPE / LEVEL	Base to Ground Floor	Ground to Roof Floor	Roof to Roof Top Floor
C1			
	10"x10"	10"x10"	
	6-16mmØ main bar	6-16mmØ main bar	
	10mmØ@4"C/C ties at Lo	10mmØ@4"C/C ties at Lo	
	10mmØ@6"C/C ties at L1	10mmØ@6"C/C ties at L1	
PC			
			10"x10"
			4-16mmØ main bar
			10mmØ@4"C/C ties at Lo
			10mmØ@6"C/C ties at L1

BEAM SCHEDULE

TYPE	SIZE	TOP REINFORCEMENT			BOTTOM REINFORCEMENT			TORSION (MIDDLE)	STIRRUPS		REMARKS
		LEFT	CENTER	RIGHT	LEFT	CENTER	RIGHT		AT SUPPORT	AT MID	
GB1	10"x10"	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	—	10mmØ@4"C/C	10mmØ@6"C/C	
GB2	10"x12"	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	—	10mmØ@4"C/C	10mmØ@6"C/C	
RB1	10"x10"	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	—	10mmØ@4"C/C	10mmØ@6"C/C	
RB2	10"x12"	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	3-16mmØ	—	10mmØ@4"C/C	10mmØ@6"C/C	

Note

-All seismic hooks (135°) should be care for alternation.

OWNER:

PROJECT:
PROPOSED ONE STOREYED R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
FOOTING,BEAM,COLUMN SCHEDULE

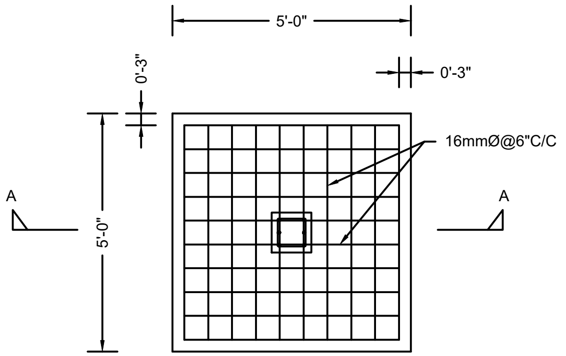
SCALE	AS SHOWN
DATE	

SHEET NO.
SD13

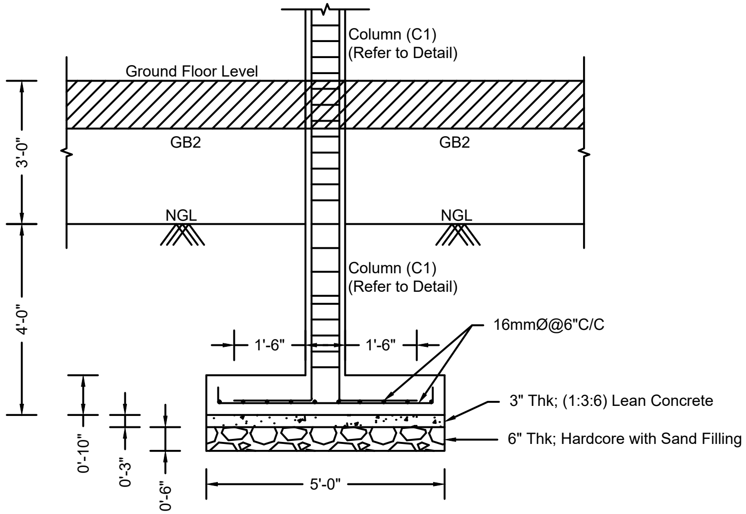
RSE (STRUCTURE):

RSE (CONSTRUCTION):

OWNER:

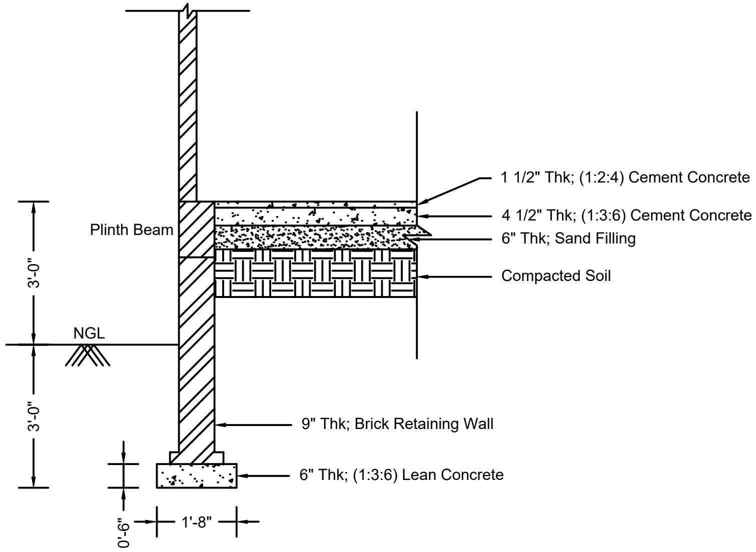


FOOTING PLAN

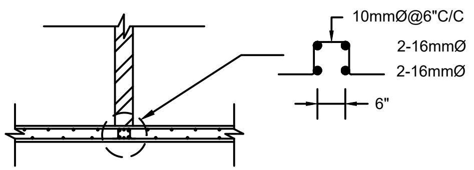


SECTION A-A

FOOTING DETAIL (GRID C2) (F3 & C1)



BRICK RETAINING WALL DETAIL

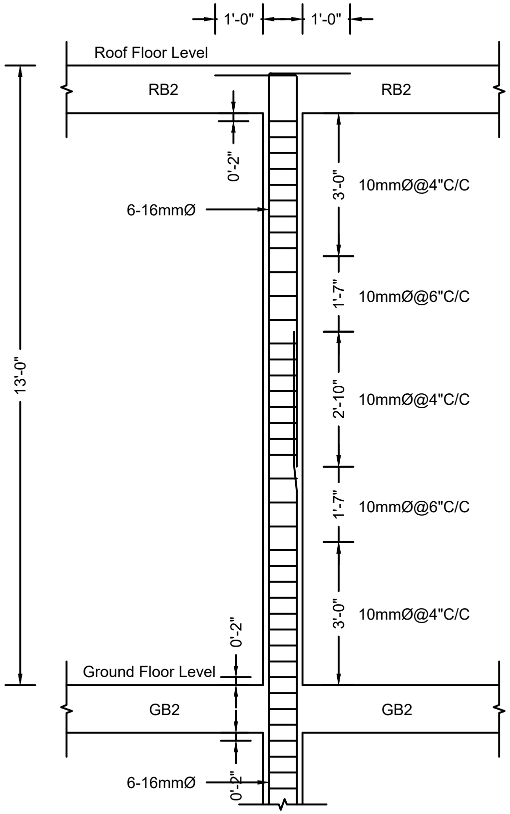


Note: Slab additional rebar go throughout to beam support.

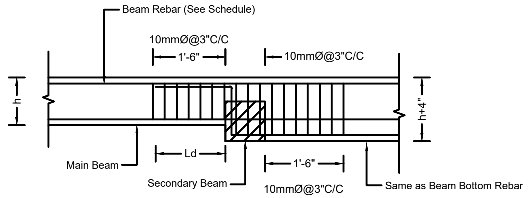
SLAB ADDITIONAL REBAR DETAIL UNDER BRICK WALL

Scale :

OWNER:	
PROJECT: PROPOSED ONE STOREYED R.C.C BUILDING	
BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW
SUBJECT: FOOTING,R-WALL DETAIL	
SCALE	
DATE	AS SHOWN
SHEET NO. SD14	
RSE (STRUCTURE):	
RSE (CONSTRUCTION):	
OWNER:	



COLUMN DETAIL (C1)

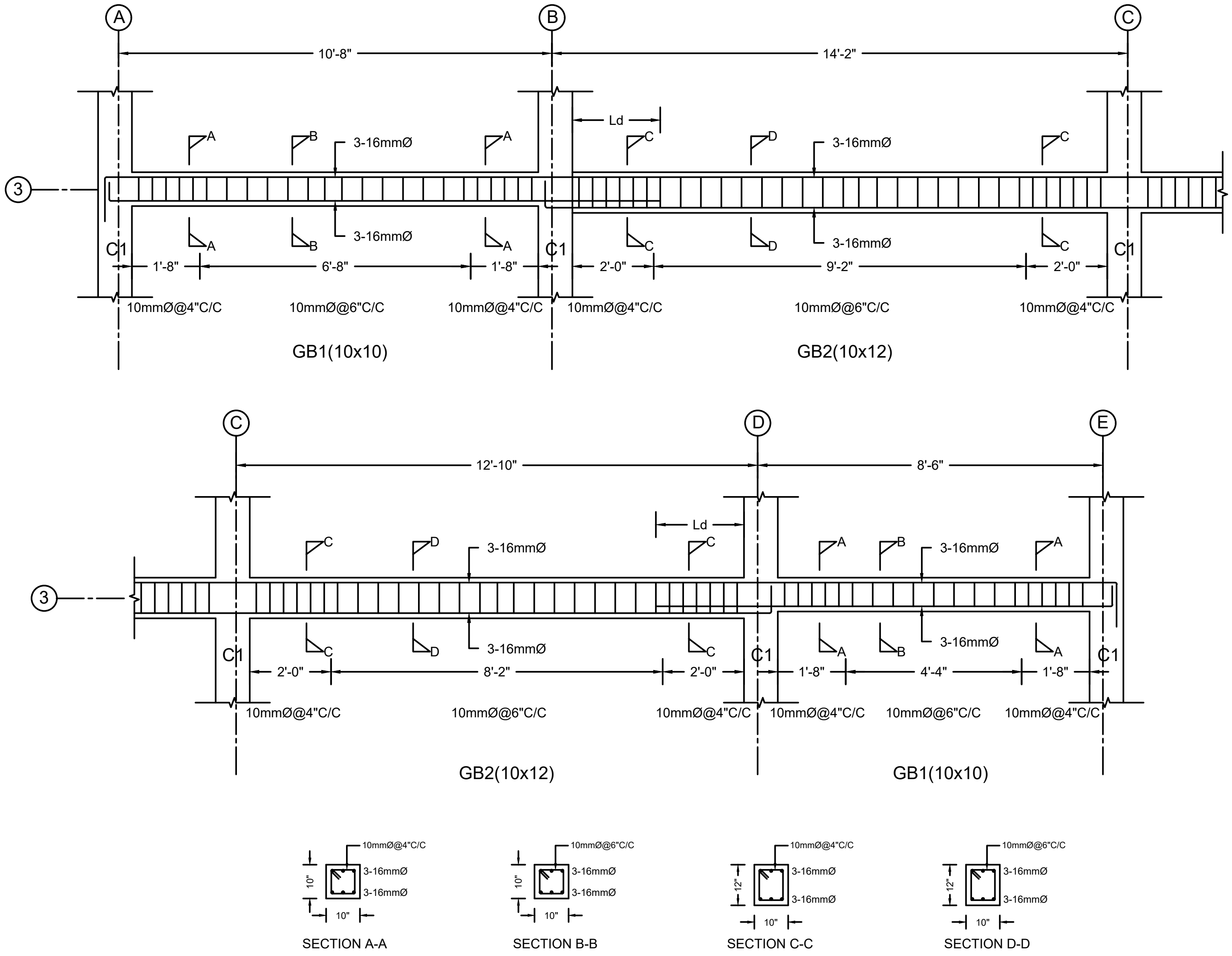


BEAM BOTTOM DROP DETAIL

Scale :

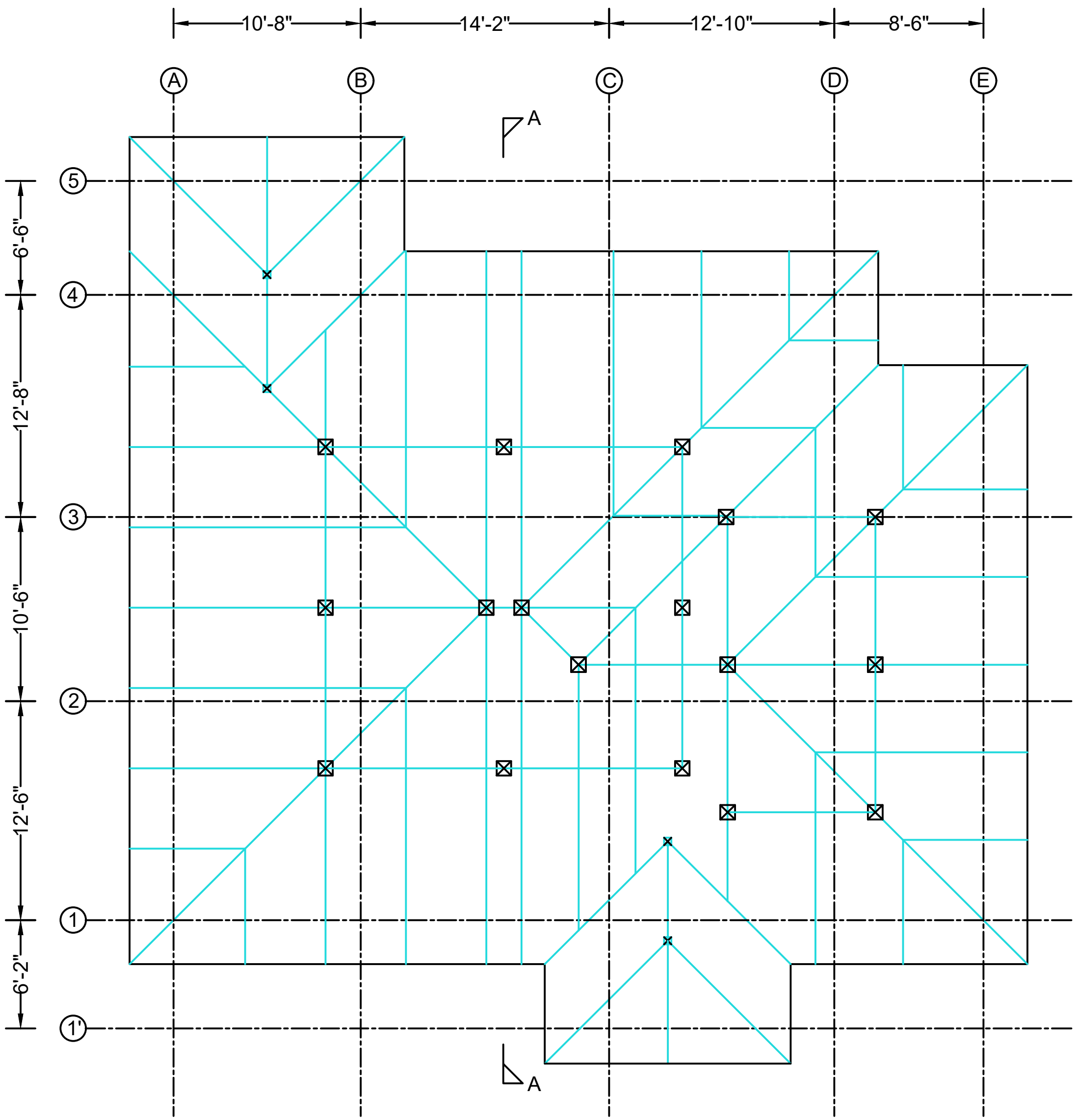
0' 2' 4' 6' 8' 10'

OWNER:	
PROJECT: PROPOSED ONE STOREYED R.C.C BUILDING	
BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW
SUBJECT: COLUMN DETAIL	
SCALE	
DATE	AS SHOWN
SHEET NO. SD15	
RSE (STRUCTURE):	
RSE (CONSTRUCTION):	
OWNER:	



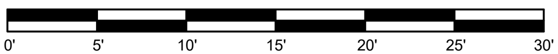
BEAM DETAIL (GRID 3)

OWNER:	
PROJECT: PROPOSED ONE STOREYED R.C.C BUILDING	
BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW
SUBJECT: BEAM DETAIL	
SCALE	
DATE	AS SHOWN
SHEET NO. SD16	
RSE (STRUCTURE):	
RSE (CONSTRUCTION):	
OWNER:	



RAFTER LAYOUT PLAN

Scale :



Note

-Rafters are 4"x2" hollow section with 1.4mm thickness.
-Rafters number and position as shown in drawing.
-Purlins are 3"x2" hollow section with 1.2mm thickness.
-Purlin spacing should not be greater than 2ft.
-All joints are welding.

OWNER:

PROJECT:
PROPOSED ONE STOREYED R.C.C BUILDING

BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW

SUBJECT:
RAFTER LAYOUT PLAN

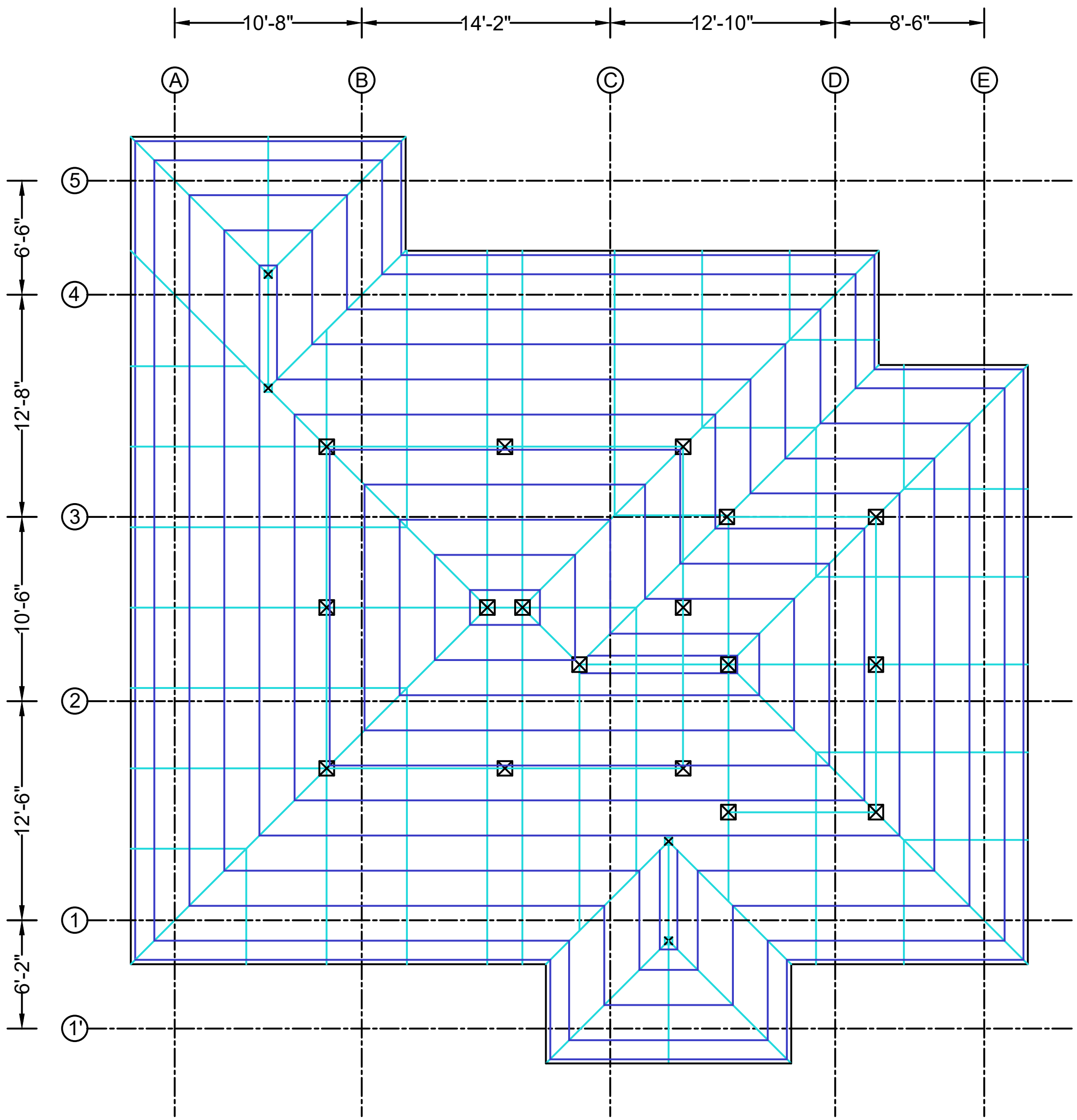
SCALE	AS SHOWN
DATE	

SHEET NO.
SD17

RSE (STRUCTURE):

RSE (CONSTRUCTION):

OWNER:



PURLIN LAYOUT PLAN

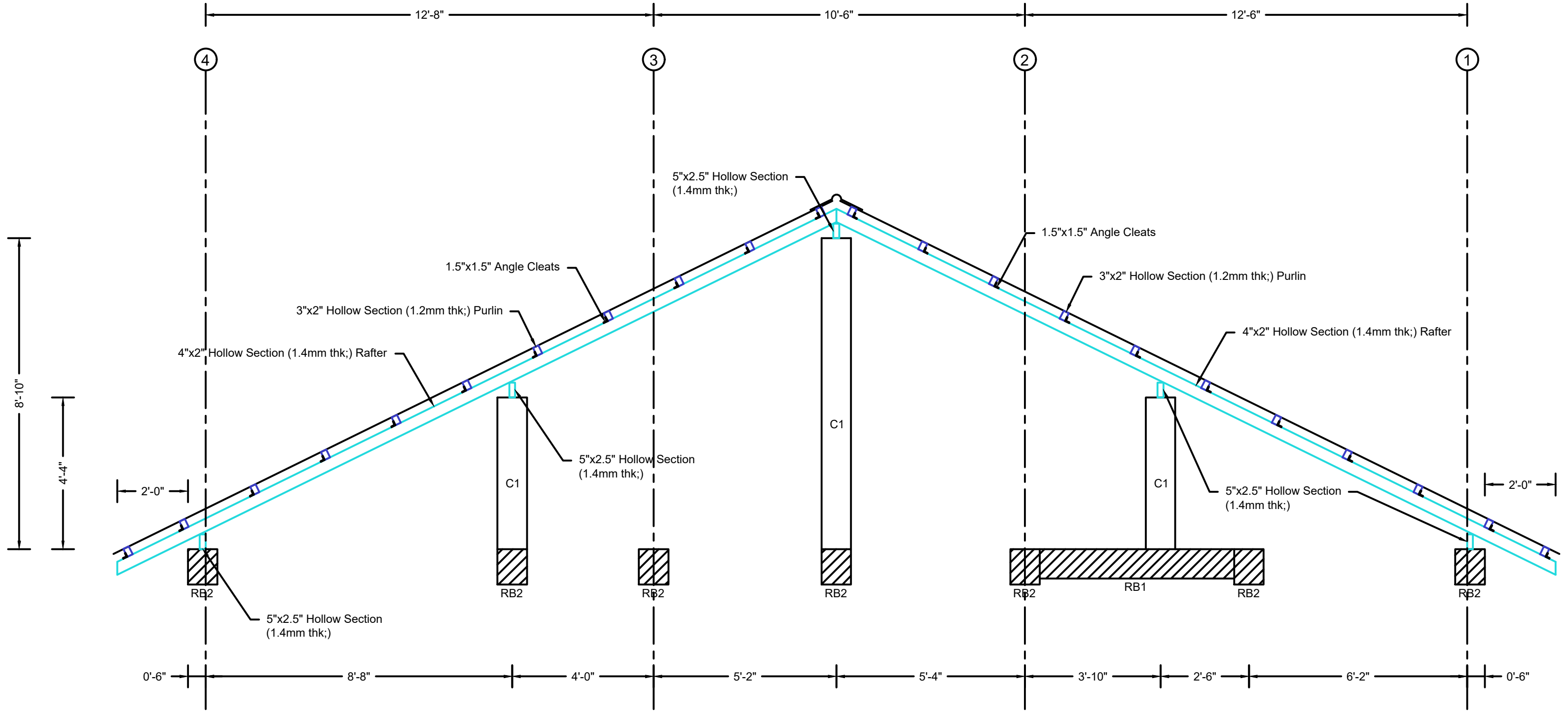
Scale :

0' 5' 10' 15' 20' 25' 30'

Note

-Rafters are 4"x2" hollow section with 1.4mm thickness.
-Rafters number and position as shown in drawing.
-Purlins are 3"x2" hollow section with 1.2mm thickness.
-Purlin spacing should not be greater than 2ft.
-All joints are welding.

OWNER:	
PROJECT: PROPOSED ONE STOREYED R.C.C BUILDING	
BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW
SUBJECT: PURLIN LAYOUT PLAN	
SCALE	AS SHOWN
DATE	
SHEET NO. SD18	
RSE (STRUCTURE):	
RSE (CONSTRUCTION):	
OWNER:	



ROOF SECTION A-A

NOTE

Roof slope must be see in architectural drawing.

Scale :

0' 2' 4' 6' 8' 10'

OWNER:	
PROJECT: PROPOSED ONE STOREYED R.C.C BUILDING	
BLOCK NO.	
LOT NO.	
TOWNSHIP:	NAYPYITAW
SUBJECT: ROOF SECTION	
SCALE	
DATE	AS SHOWN
SHEET NO. SD19	
RSE (STRUCTURE):	
RSE (CONSTRUCTION):	
OWNER:	